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Marks	0.00/41.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 1.00

Flag question

Jane is a 35-year-old lady with treatment resistant schizophrenia. She is being admitted to the inpatient unit for clozapine titration and is particular concerned about constipation after reading some stories in the media about life threatening bowel obstruction. She has asked her doctor if there are any particular risk factors associated with clozapine-induced constipation. Which of the following factors have empirical evidence for being associated with an increased risk of clozapine-induced constipation?

Select one:

- Diagnosis of schizophrenia
- Smoking
- Female gender
- Inpatient mental health treatment
- Decreasing age

Your answer is incorrect.

A recent meta-analysis has studied the currently available evidence for risk factors in clozapine related constipation. Meta-regression identified that significantly higher ($p = 0.02$) rates of constipation were observed for those treated in inpatient versus outpatient or mixed settings. In meta-regression analyses age, sex, diagnosis, percentage of smokers, duration of treatment, clozapine dose, and plasma clozapine and norclozapine concentrations surprisingly did not predict constipation. (Ref: Shirazi A, Stubbs B, Gomez L, Moore S, Gaughran F, Flanagan RJ, MacCabe JH, Lally J. Prevalence and predictors of clozapine-associated constipation: a systematic review and meta-analysis. International journal of molecular sciences. 2016 Jun 2;17(6):863)

The correct answer is: Inpatient mental health treatment

Question **2**

Not answered

Marked out of 1.00

Flag question

Mr Steven Marks is a 42-year-old man with a history of opiate dependence who has recently engaged with his local addiction service and wishes to cease his use of illicit opioids. He has tried this in the past and has had difficulty with light-headedness and episodes of fainting when prescribed certain medications. Which of the following medications is most likely to cause these effects?

Select one:

- Methadone
- Naltrexone
- Clonidine
- Lofexidine
- Buprenorphine

Your answer is incorrect.

Clonidine is associated with light-headedness and troublesome hypotension leading to episodes of fainting. Lofexidine is not associated with troublesome hypotension.

The correct answer is: Lofexidine

Question **3**

Not answered

Marked out of 1.00

Flag question

Jane is a 23-year-old aspiring actor. She has been to see her GP for depression and wants to try an antidepressant medication. She is very concerned about the potential side effect of weight gain. Which one of the following antidepressants has the greatest risk of weight gain, and is therefore best avoided in Jane's case?

Select one:

- Fluoxetine
- Sertraline
- Mirtazapine
- Venlafaxine
- Escitalopram

Your answer is incorrect.

A meta-analysis conducted found that amitriptyline, mirtazapine, and paroxetine were associated with a greater risk of weight gain. In contrast, some weight loss occurs with fluoxetine and bupropion, although the effect of fluoxetine appears to be limited to the acute phase of treatment. Other compounds have no transient or negligible effect on body weight in the short term. (Ref: Serretti A, Mandelli L. Antidepressants and body weight: a comprehensive review and meta-analysis. Journal of Clinical Psychiatry. 2010;71(10):1259-72).

The correct answer is: Mirtazapine

Question **4**

Not answered

Marked out of 1.00

Flag question

Mrs. Watkins has recently started sertraline and has noticed she has been feeling dizzy and lethargic. Her GP checks her sodium and finds that it is 128. Which of the following is a risk factor for antidepressant-induced hyponatraemia?

Select one:

- Normal renal function
- Patients also taking diuretics
- History of hypernatremia

- Male gender
- Young age

Your answer is incorrect.

High risk factors for drug induced hyponatremia include extreme old age (>80 years), female gender, history of hyponatremia or low baseline Na concentration, co-therapy with other drugs associated with hyponatremia (diuretics, NSAIDs, carbamazepine, ACE inhibitors, calcium antagonists), reduced renal function, and medical co-morbidities. (Ref: Taylor, D., Paton, C. and Kapur, S. The Maudsley prescribing guidelines in psychiatry. 12th ed. Chichester: John Wiley & Sons, pp.316) High risk factors for drug induced hyponatremia include extreme old age (>80 years), female gender, history of hyponatremia or low baseline Na concentration, co-therapy with other drugs associated with hyponatremia (diuretics, NSAIDs, carbamazepine, ACE inhibitors, calcium antagonists), reduced renal function, and medical co-morbidities. (Ref: Taylor, D., Paton, C. and Kapur, S. The Maudsley prescribing guidelines in psychiatry. 12th ed. Chichester: John Wiley & Sons, pp.316) High risk factors for drug induced hyponatremia include extreme old age (>80 years), female gender, history of hyponatremia or low baseline Na concentration, co-therapy with other drugs associated with hyponatremia (diuretics, NSAIDs, carbamazepine, ACE inhibitors, calcium antagonists), reduced renal function, and medical co-morbidities. (Ref: Taylor, D., Paton, C. and Kapur, S. The Maudsley prescribing guidelines in psychiatry. 12th ed. Chichester: John Wiley & Sons, pp.316)

The correct answer is: Patients also taking diuretics

Question **5**

Not answered

Marked out of 1.00

Flag question

Psychiatric medications are commonly associated with weight changes, with some causing significant increases in appetite and weight which patients can find troublesome. Which of the following medications is most associated with weight gain?

Select one:

- Vortioxetine
- Agomelatine
- Modafinil
- Pregabalin
- Methylphenidate

Your answer is incorrect.

Pregabalin is the only medication of the above commonly associated with weight gain. Methylphenidate and Modafinil are stimulants and are more likely to be associated with weight loss.

The correct answer is: Pregabalin

Question **6**

Not answered

Marked out of 1.00

Mr Ivor Staines is a 25-year-old man with a diagnosis of paranoid schizophrenia. He has struggled to take oral medication and has been started on depot Haloperidol that has resolved his psychotic symptoms. However, he has been troubled by stiffness when moving and slowed physical reactions. Which of the following treatments would be most appropriate

Flag question

to treat these symptoms?

Select one:

- Propranolol
- Valbenazine
- Topiramate
- Procyclidine
- Diazepam

Your answer is incorrect.

Procyclidine is often used to treat extrapyramidal side-effects, particularly those which are Parkinsonian in character. Valbenazine, vesicular monoamine transporter 2 (VMAT2) Inhibitor, is used to treat tardive dyskinesia.

The correct answer is: Procyclidine

Question **7**

Not answered

Marked out of 1.00

Flag question

Lamotrigine is a mood stabilising medication that appears to have beneficial effects for patients experiencing recurrent depressive episodes, either in the context of Type II bipolar disorder or recurrent depression. Which of the following is a rare but serious potential side-effect of Lamotrigine?

Select one:

- Hypothyroidism
- Steven-Johnson's syndrome
- Agranulocytosis
- Poly-cystic ovarian syndrome
- Renal impairment

Your answer is incorrect.

Lamotrigine has been associated with Steven-Johnson's syndrome, hence needs to be gradually titrated up with patients being aware of the need to inform medical services if they develop a new rash, particularly one affecting mucosal surfaces. Agranulocytosis can occur with Clozapine and PCOS in Valproate. Hypothyroidism and renal impairment are associated with Lithium use.

The correct answer is: Steven-Johnson's syndrome

Question **8**

Not answered

Marked out of 1.00

Flag question

Mr Jason Hardy is a 28-year-old man with treatment resistant social phobia. He is started on Phenelzine to try and reduce his symptoms. Which of the following does he need to be aware of prior to starting this medication?

Select one:

- To avoid fermented food and drink
- To ensure he takes it at exactly the same time each day

- To inform the team if he develops a new rash
- To avoid concurrent use of NSAIDs
- To seek medical assistance if he develops a fever

Your answer is incorrect.

Monoamine oxidase inhibitors act by inhibiting the activity of monoamine oxidase, thus preventing the breakdown of monoamine neurotransmitters and thereby increasing their availability. Patients taking these medications must avoid eating foods with high levels of tyramine such as animal liver and fermented substances, such as alcoholic beverages and aged cheeses. Excessive concentrations of tyramine in blood plasma can lead to hypertensive crisis by increasing the release of noradrenaline, which causes blood vessels to constrict by activating alpha-1 adrenergic receptors. Ordinarily, MAO-A would metabolise the tyramine and excess noradrenaline.

The correct answer is: To avoid fermented food and drink

Question **9**

Not answered

Marked out of 1.00

Flag question

When treating pregnant women with mental health difficulties, the benefits of medications must be carefully balanced against the potential risks involved. Which of the following medications is felt to be most risky for a developing foetus?

Select one:

- Carbamazepine
- Diazepam
- Olanzapine
- Lamotrigine
- Valproate

Your answer is incorrect.

Valproate can result in congenital malformations in up to 10% of cases and a well-known valproate syndrome in up to 40% of children exposed to it in utero. This condition is associated with developmental and cognitive difficulties. NICE recommends that valproate is not used in women of child-bearing age unless there is no other alternative.

The correct answer is: Valproate

Question **10**

Not answered

Marked out of 1.00

Flag question

Clozapine is used in patients with treatment resistant schizophrenia as it can benefit patients who do not respond to other medications. However, it can have significant side-effects. Which of the following are most commonly seen as a side-effect of Clozapine?

Select one:

- Bradycardia
- Muscular pain
- Diarrhoea
-

Weight gain

Agranulocytosis

Your answer is incorrect.

Weight gain is a very common side effect of Clozapine. Constipation and tachycardia are also commonly seen. Agranulocytosis is a serious potential side-effect but is rare, only occurring in around 0.4% of patients.

The correct answer is: Weight gain

Question **11**

Not answered

Marked out of 1.00

Flag question

Mr James is a 52-year-old man who has been clinically stable on lithium therapy for many years. He visits his GP for his routine yearly health check and is found to have abnormalities on his ECG. Which of the following ECG abnormalities have not been associated with lithium therapy?

Select one:

- Higher QT dispersion ratio
- Sinus node dysfunction
- Wolff Parkinson White syndrome
- T wave flattening
- PR prolongation

Your answer is incorrect.

Sinus node dysfunction, higher QT dispersion ratio, T wave flattening and PR prolongation (amongst other abnormalities) have been identified in patients taking lithium therapy. Wolff-Parkinson-White (WPW) syndrome is a congenital condition involving abnormal conduction between the atria and the ventricles, which can provide a pathway for a re-entrant tachycardia circuit, in association with supraventricular tachycardia (SVT). It is congenital and therefore not related to lithium therapy. (Ref: Mehta N, Vannozzi R. Lithium-induced electrocardiographic changes: A complete review. Clinical cardiology. 2017 Dec 1.Di Biase, L & Walsh, E. Epidemiology, clinical manifestations and diagnosis of the Wolff Parkinson White Syndrome. UptoDate. 2018)

The correct answer is: Wolff Parkinson White syndrome

Question **12**

Not answered

Marked out of 1.00

Flag question

Benzodiazepines are important class of medications that have application as sedatives and hypnotics. Which of the following best describes their mechanism of action?

Select one:

- GABA-B agonist
- GABA-A positive allosteric modulator
- GABA-A antagonist
- GABA-B positive allosteric modulator
- GABA-A agonist

Your answer is incorrect.

Benzodiazepines work by increasing the efficiency of a natural brain chemical, GABA, to decrease the excitability of neurones. GABA controls the excitability of neurons by binding to the GABA-A receptor. Benzodiazepines bind to the GABA-A receptor and acts as a positive allosteric modulator by increasing the total conduction of chloride ions across the neuronal cell membrane when GABA is already bound to its receptor. This increased chloride ion influx hyperpolarizes the neurone's membrane potential. As a result, the difference between resting potential and threshold potential is increased and firing is less likely.

The correct answer is: GABA-A positive allosteric modulator

Question **13**

Not answered

Marked out of 1.00

Flag question

Michael is a 27-year-old insurance worker who regularly uses cocaine. He has recently started dating a biology student who informs him about the neurochemical effects of the drug. Changes in which of the following mechanisms are thought to be responsible for the euphoric effects of cocaine?

Select one:

- Increased concentration of glutamate
- Increased levels of noradrenaline
- Sodium channel blockade
- Inhibition of dopamine reuptake
- Reduced availability of dopamine

Your answer is incorrect.

The rewarding effects of stimulants such as cocaine are related to increases in dopamine concentrations in limbic regions that is due either to dopamine transporter blockade direct occupancy of dopamine D2 receptors. Though PET studies strongly indicate a role for dopamine in cocaine use, drugs that modify dopamine transmission have not been useful in the treatment of cocaine use. Dopamine receptor subtype D1 is implicated in the conditioned responses to cocaine. Ref: Volkow ND, Wang GJ, Fowler JS, Tomasi D, Telang F, Baler R. Addiction: decreased reward sensitivity and increased expectation sensitivity conspire to overwhelm the brain's control circuit. Bioessays. 2010 Sep;32(9):748-55.

The correct answer is: Inhibition of dopamine reuptake

Question **14**

Not answered

Marked out of 1.00

Flag question

Mrs Stephanie Connor is a 73-year-old woman who has recently been diagnosed with Parkinson‘s disease and has been started on Carbidopa. She is a retired biology teacher and is curious about how the medication works. Which of the following best describes its mechanism of action?

Select one:

- Inhibits DOPA decarboxylase
- Dopamine antagonist
- Dopamine partial agonist
-

Increases metabolism of L-dopa (Levodopa) in the CNS

Prevents breakdown of dopamine in CNS

Your answer is incorrect.

The objective of treatment in Parkinson's disease is to increase the levels of dopamine in the CNS of patients. L-dopa acts as a precursor of dopamine and can cross the blood-brain barrier, unlike dopamine. Carbidopa inhibits DOPA decarboxylase, which metabolises L-dopa into dopamine, outside the CNS and thus increases the amount of L-DOPA, which is available to cross into the CNS and then become dopamine. Carbidopa does not cross the blood-brain barrier and so does not interfere with L-dopa metabolism in the CNS.

The correct answer is: Inhibits DOPA decarboxylase

Question **15**

Not answered

Marked out of 1.00

Flag question

James is a 22-year-old man who injects heroin daily. He has attended his local substance misuse service and is discussing possible maintenance or substitution therapies. Which of the following medicine is correctly matched with its mechanism of action?

Select one:

Buprenorphine – pure opioid agonist

Buprenorphine – pure opioid antagonist

Naltrexone – partial opioid agonist

Methadone – pure opioid agonist

Methadone – partial opioid agonist

Your answer is incorrect.

Methadone is a pure opioid agonist. Buprenorphine is a partial opioid agonist. Naltrexone is a competitive opioid antagonist. (Ref: Standiford Helm II, Trescot AM, Colson J, Sehgal N, Silverman S. Opioid antagonists, partial agonists, and agonists/antagonists: the role of office-based detoxification. Pain Physician. 2008 Mar;11:225-35).

The correct answer is: Methadone – pure opioid agonist

Question **16**

Not answered

Marked out of 1.00

Flag question

Which of the following neurotransmitters is used by the neurons within the locus coeruleus as their primary output?

Select one:

Noradrenaline

GABA

Dopamine

Glutamate

Serotonin

Your answer is incorrect.

The locus coeruleus ("blue site“) is the principal nucleus for brain synthesis of noradrenaline. The locus coeruleus neurons project to the relay stations in the thalamus, hypothalamus and basal forebrain, as well as further along to innervate the entire cerebral cortex directly (Ref: Barbara E. Jones, in Handbook of Clinical Neurology, 2011)

The correct answer is: Noradrenaline

Question **17**

Not answered

Marked out of 1.00

Flag question

Atomoxetine is a medication commonly used in the treatment of ADHD in children. Which of the following options best describes its mechanism of action?

Select one:

- Serotonin reuptake inhibitor
- Dopamine reuptake inhibitor
- Glutamate agonist
- Noradrenaline reuptake inhibitor
- GABA antagonist

Your answer is incorrect.

Atomoxetine is a noradrenaline reuptake inhibitor. Methylphenidate mainly works as a dopamine reuptake inhibitor, although it also has noradrenaline reuptake inhibition actions.

The correct answer is: Noradrenaline reuptake inhibitor

Question **18**

Not answered

Marked out of 1.00

Flag question

Mrs. Henry is a 60-year-old woman with a history of myocardial infarction and depression. She is taking both sertraline and aspirin. Which of the following is correct regarding the effect of selective serotonin reuptake inhibitors (SSRIs) on potential risk of gastrointestinal (GI) bleeding?

Select one:

- Increased levels of serotonin interfere with vasoconstriction
- SSRIs inhibit the transporter responsible for uptake of serotonin into platelets
- SSRIs cause increased level of gut serotonin resulting in platelet aggregation
- Increased levels of serotonin interfere with conversion of prothrombin to thrombin
- SSRIs decrease gastric secretion and act as an irritant to the gastric mucosa

Your answer is incorrect.

Serotonin is released from platelets in response to vascular injury and promotes vasoconstriction and morphological changes in platelets that lead to aggregation. SSRIs inhibit the serotonin transporter, which is responsible for the uptake of serotonin into platelets. SSRIs therefore deplete platelet serotonin and lead to reduced ability to form clots. SSRIs also increase gastric acid secretion and can be an irritant to the gastric mucosa. Patients who are taking SSRIs are at significant risk of GI bleeding compared to age and sex matched controls. Co-prescription of low dose aspirin doubles the risk associated with SSRIs alone.

(Ref: Taylor, D., Paton, C. and Kapur, S. (n.d.). The Maudsley prescribing guidelines in psychiatry. 12th ed. Chichester: John Wiley & Sons, pp.328-329)

The correct answer is: SSRIs inhibit the transporter responsible for uptake of serotonin into platelets

Question **19**

Not answered

Marked out of 1.00

Flag question

Gamma hydroxybutric acid (GHB) acts on

Select one:

- Monoamine oxidase
- GABA-B
- GABA-A
- Glutamate
- COMT

Your answer is incorrect.

Gamma-hydroxybutyric acid (GHB), also known as 4-hydroxybutanoic acid, is a naturally occurring neurotransmitter and a psychoactive drug. It is a precursor to GABA, glutamate, and glycine in certain brain areas, and it acts on the GHB receptor and it is a weak agonist at the GABA-B receptor.

The correct answer is: GABA-B

Question **20**

Not answered

Marked out of 1.00

Flag question

Some medications interact with one another by increasing or decreasing their serum concentration via a variety of mechanisms. Which of the following combinations of medication is most associated with significant changes in the serum concentrations of one or both drugs?

Select one:

- Valproate and Lamotrigine
- Amitriptyline and tramadol
- Clozapine and lithium
- Citalopram and aspirin
- Risperidone and smoking

Your answer is incorrect.

Valproate inhibits the clearance of lamotrigine by approximately 30% at a dose of 125mg/day and by 50% at doses of 250 mg and higher. Concurrent use of amitriptyline and tramadol increases the risk of seizure activity. Clozapine and lithium can be associated with an increased risk of neurotoxicity even within therapeutic serum concentrations. Clozapine serum levels are significantly reduced by tobacco use due to induction of CYP1A2. There is a possible small effect on Risperidone metabolism but it is not considered to be clinically relevant. Citalopram inhibits serotonin reuptake; in the periphery this impairs the ability of platelets to store serotonin, which they use to enhance their aggregation. This increases the

risk of bleeding, particularly when combined with other medications, which impair platelet function.

The correct answer is: Valproate and Lamotrigine

Question **21**

Not answered

Marked out of 1.00

Flag question

Some medications are associated with a significant withdrawal syndrome, which can be very distressing for some patients. Which of the following medications is most associated with "waves“ of withdrawal symptoms?

Select one:

- Vortioxetine
- Sertraline
- Mirtazapine
- Venlafaxine
- Fluoxetine

Your answer is incorrect.

Venlafaxine is associated with a severe withdrawal syndrome with some patients describing symptoms coming in waves. The relatively short half-life of Venlafaxine of around 5 hours appears to contribute towards the severity of the syndrome. People stopping venlafaxine commonly experience discontinuation symptoms such as dysphoria, headaches, nausea, irritability, emotional lability, sensation of electric shocks and sleep disturbance. Venlafaxine has a higher rate of moderate to severe discontinuation symptoms relative to other antidepressants.

The correct answer is: Venlafaxine

Question **22**

Not answered

Marked out of 1.00

Flag question

Sedative medications can often result in troublesome hangover effects when used as hypnotics. Which of the following sedatives is the shortest acting?

Select one:

- Temazepam
- Diazepam
- Zolpidem
- Zopiclone
- Zaleplon

Your answer is incorrect.

Zaleplon has a half-life of around 1 hour. Zolpidem and Zopiclone have half-lives of 2.4 hours and 5 hours respectively. The half-life of Diazepam and its active metabolite can be up to 100 hours, while Temazepam has a half-life of around 10 hours.

The correct answer is: Zaleplon

Question **23**

Not answered

Marked out of 1.00

Flag question

Bobby Milheim was diagnosed with depression three months ago, and has not so far been helped by Mirtazapine. He wishes to switch to an SSRI but wants to know how long the drug and its metabolites are likely to remain in his body for. He asks which of the following has the shortest half-life:

Select one:

- Fluoxetine
- Fluvoxamine
- Paroxetine
- Escitalopram
- Sertraline

Your answer is incorrect.

Fluvoxamine, which has a half-life of 17-22 hours. Sertraline has a half-life of 22-36 hours, Paroxetine has a half-life of about 24 hours, Fluoxetine has a half-life of 4-6 days, Escitalopram has a half-life of about 30 hours Fluvoxamine, which has a half-life of 17-22 hours. Sertraline has a half-life of 22-36 hours, Paroxetine has a half-life of about 24 hours, Fluoxetine has a half-life of 4-6 days, Escitalopram has a half-life of about 30 hours

The correct answer is: Fluvoxamine

Question **24**

Not answered

Marked out of 1.00

Flag question

Bernadette Brown is 79 years old. Her family have been concerned about her increasing struggles with memory and general functioning. She‘s not as quick as she used to be, and previous tasks (such as sorting out the cutlery draws) seem to be a particular challenge for her now. You wish to perform a neuropsychological test to appraise Mrs Brown‘s visual attention and task-switching skills. Which of the following tests do you choose?

Select one:

- Rey-Osterrieth Complex Figure
- Trail Making test
- Stroop task
- Boston Naming test
- Tower of London test

Your answer is incorrect.

In the Trail Making test the subject is invited to connect a set of 25 dots as quickly as possible, whilst maintaining accuracy. It provides information about visual search speed, scanning, speed of processing, metal flexibility and executive functioning.

The correct answer is: Trail Making test

Question **25**

Not answered

Mr Nicholas Evans is a 72 year old man who has been increasingly disinhibited and garrulous in social situations despite having been rather reserved before. His wife has noticed that he struggles to get himself ready in the mornings. He is referred to for a

Marked out of 1.00

Flag question

particular psychological test. Which test is most suitable to identify his problems?

Select one:

- Bender-Gestalt Test
- Wisconsin card sorting test
- Raven‘s progressive matrices
- National adult reading test
- Dot cancellation test

Your answer is incorrect.

The Wisconsin card sorting test is used to assess frontal lobe functions, particularly set-shifting.

The correct answer is: Wisconsin card sorting test

Question **26**

Not answered

Marked out of 1.00

Flag question

Mrs Cram understands that her father, who was dependent on alcohol for many decades, has developed Korsakoff syndrome. She finds that his memory is significantly affected. Her mother developed Alzheimer‘s and so she is familiar with having a parent whose memory is impaired. Mrs Cram wishes to know what memory deficit is the hallmark of Korsakoff. Which of the following is correct?

Select one:

- Declarative memory
- Procedural memory
- Implicit memory
- Autobiographical memory
- Anterograde memory

Your answer is incorrect.

Anterograde amnesia is the hallmark of Korsakoff, but there‘s also retrograde amnesia and amnesia of fixation (loss of immediate memory). Confabulation occurs frequently to make up for gaps in the person‘s memory.

The correct answer is: Anterograde memory

Question **27**

Not answered

Marked out of 1.00

Flag question

Which of the following is NOT a component of Gerstmann syndrome?

Select one:

- Acalculia
- Finger agnosia
- Dysgraphia
- Constructional apraxia

- Left/right disorientation

Your answer is incorrect.

Damage to the inferior parietal lobule of the dominant hemisphere results in Gerstmann's syndrome. It is characterized by four primary symptoms: a) Dysgraphia/agraphia: deficiency in the ability to write b) Dyscalculia/acalculia: difficulty in learning or comprehending mathematics c) Finger agnosia: inability to distinguish the fingers on the hand d) Left-right disorientation

The correct answer is: Constructional apraxia

Question **28**

Not answered

Marked out of 1.00

Flag question

Mr Williams is a 70 year old man who has presented to his GP with difficulty co-ordinating the movements of his right arm. Which of the following neurodegenerative disorders is most likely to be responsible?

Select one:

- Corticobasal degeneration
- Lewy Body Dementia
- Bulbar palsy
- Progressive supranuclear palsy
- Wernicke‘s encephalopathy

Your answer is incorrect.

Corticobasal degeneration is a rare, progressive neurodegenerative disease involving the cerebral cortex and the basal ganglia. The movement symptoms typically appear asymmetrically and include apraxia, aphasia, parkinsonism and alien hand syndrome.

The correct answer is: Corticobasal degeneration

Question **29**

Not answered

Marked out of 1.00

Flag question

Which of the following is one of the brain areas that is affected very early in Alzheimer‘s disease?

Select one:

- Entorhinal cortex
- Cerebellum
- Mid brain raphe nucleus
- Fusiform gyrus
- Angular gyrus

Your answer is incorrect.

"At first, Alzheimer‘s disease typically destroys neurons and their connections in parts of the brain involved in memory, including the entorhinal cortex and hippocampus. It later affects

areas in the cerebral cortex responsible for language, reasoning, and social behavior. Eventually, many other areas of the brain are damaged“. Ref: Excerpt from https://www.nia.nih.gov/health/what-happens-brain-alzheimers-disease

The correct answer is: Entorhinal cortex

Question **30**

Not answered

Marked out of 1.00

Flag question

Lily Flanders is 89 years old and lives in a care home. She has had gradual cognitive impairment for the past decade, and over the past 6 months her behaviour has become more unpredictable: sometimes she can be extremely friendly and over-familiar, at other times she‘s irritable, shouting at the staff. She experiences a fall with minor head injury, and whilst admitted to the ward an MRI of her brain is requested. Which areas of the brain are most likely to be affected in relation to Mrs Flander‘s dementia and behavioural disturbance, as shown by MRI investigation?

Select one:

- Parietal lobes
- Basal ganglia
- Fronto-temporal region
- Hypothalamus
- Cerebellar vermis

Your answer is incorrect.

Fronto-temporal region, whose functioning is responsible for cognition and behaviour, is likely to be atrophied in cases of dementia with behavioural disturbance.

The correct answer is: Fronto-temporal region

Question **31**

Not answered

Marked out of 1.00

Flag question

Which of the following diseases has a purely maternal inheritance?

Select one:

- Angelman's syndrome
- Rett's syndrome
- Holoprosencephaly
- Leber's hereditary optic neuropathy
- Turner's syndrome

Your answer is incorrect.

Leber's hereditary optic neuropathy is inherited via mitochondrial DNA and so is purely inherited via the maternal line.

The correct answer is: Leber's hereditary optic neuropathy

Question **32**

Mrs Pamela West is a 75-year-old woman who has presented to her GP with tremor and

Not answered

Marked out of 1.00

Flag question

difficulty walking. Her husband has noted that her face often does not respond when she is told about happy or sad news. Which of the following is associated with this condition?

Select one:

- Tau protein
- Amyloid precursor protein
- LRRK2 mutation
- Pre-senilin 1 mutations
- Ubiquitin

Your answer is incorrect.

Parkinson's disease is characterised by the cell death of dopaminergic neurones in the basal ganglia, Lewy bodies (alpha synuclein). The LRRK2 mutation is associated with some cases of Parkinson's disease.

The correct answer is: LRRK2 mutation

Question **33**

Not answered

Marked out of 1.00

Flag question

A chromosome whose centromere is in the middle is said to be:

Select one:

- Holocentric
- Acrocentric
- Metacentric
- Sub-metacentric
- Telocentric

Your answer is incorrect.

Metacentric chromosomes have the centromere roughly in the middle. If the chromosome arm lengths are unequal they are described as sub-metacentric. Telocentric centromeres are at the terminal end of the chromosome arms. A sub-telocentric centromere is close to the end of the chromosome arm. In holocentric chromosomes the entire chromosome acts as a centromere. If the p (short) arm is so short that it is hard to observe, but still present, then the chromosome is acrocentric.

The correct answer is: Metacentric

Question **34**

Not answered

Marked out of 1.00

Flag question

Muhamed Husain began a preoccupation with space travel in his late teens. When he left home to go to University the preoccupation intensified and took on a paranoid flavour. In his second term at University, after a period of not sleeping due to high levels of anxiety, he became convinced that alien life forms had invaded the walls of his bedroom, were continually spying on him, putting their ‘sensors' into his body and monitoring his heart-rate and breathing. After a period of hospital admission he was diagnosed with paranoid schizophrenia. He comes to see you in an outpatient clinic three years after this first admission. He's returned to his degree, and is functioning reasonably well in other aspects of

life. As part of his dissertation on genetics he's investigating copy number variations and wishes to know more about the links with schizophrenia. He makes a number of statements about what he's remembered from his reading. Which of the following statements copy number variations (CNV) in schizophrenia is not true:

Select one:

- Rare CNVs in people with schizophrenia were found to be enriched for genes that function with N-methyl-D-aspartate receptor postsynaptic signalling complexes
- Persons with schizophrenia are less likely than controls to carry genomic duplications or deletions that resulted in a fusion gene
- Most rare copy number mutations implicated in schizophrenia are unique and many arose de novo or in recent generations
- Individuals with schizophrenia are significantly more likely than unaffected persons to harbour rare gene-impacting CNVs
- Genes disrupted by CNVs in people with schizophrenia function disproportionately in signalling and neurodevelopmental processes

Your answer is incorrect.

The opposite is true (they are more likely) (Ref: Neuropsychopharm 2015;40(1):252-254)

The correct answer is: Persons with schizophrenia are less likely than controls to carry genomic duplications or deletions that resulted in a fusion gene

Question **35**

Not answered

Marked out of 1.00

Flag question

Various genes have been identified as being associated with developing Alzheimer's disease, with some predisposing to early onset illness, before the age of 65 years, and others to late onset. Which of the following genes are associated with early onset Alzheimer's disease?

Select one:

- PARK 2
- NOTCH3
- Apo E2
- Presenilin 1
- Apo E4

Your answer is incorrect.

Presenilin 1 and 2 and Amyloid Percursor Protein (APP) are associated with early onset Alzheimer's disease. Apo E4 alleles increase the likelihood of late onset Alzheimer's, whereas Apo E2 is protective. PARK2 is associated with Parkinson's disease and NOTCH3 with CADASIL.

The correct answer is: Presenilin 1

Question **36**

Trevor Wilkins has been going through a very difficult time, with the break-up of his marriage

Not answered

Marked out of 1.00

Flag question

and the recent loss of his job. He's been tempted to use alcohol as a means of alleviating his distress. His brother is dependent on alcohol, as was his father. He's concerned about the heritability of alcohol dependence, and wonders to what extent it contributes to the possibility that he too may become dependent on alcohol. Which of the following statements is true?

Select one:

- Alcohol heritability is relevant to male relatives only
- The genes implicated as the same ones as those involved in the metabolism of alcohol
- Shared environmental variance contribute to 0.8 heritability for alcohol dependence
- The genetic heritability of alcohol dependence is 80%
- There's a strong link between alcohol dependence and copy number variations of chromosome 5q13.2

Your answer is incorrect.

Genetic heritability of alcohol dependence is approximately 50%, with environmental variance contributing a smaller proportion (wide variation in studies, but overall thought to be around 10%). Genes that affect rates of alcohol metabolism (such as those carried in some people of Asian descent) are protective in the development of alcohol dependence (because they cause ill effects with the consumption of alcohol). (Ref: Psychol Med 2015;45(5):1061-1072)

The correct answer is: There's a strong link between alcohol dependence and copy number variations of chromosome 5q13.2

Question **37**

Not answered

Marked out of 1.00

Flag question

Which of the following codons signifies chain termination?

Select one:

- GUA
- AGU
- UUA
- GAA
- UGA

Your answer is incorrect.

The three stop codons are UAA, UAG, and UGA, where U is uracil, A is adenine, and G is guanine. These are also called termination codons.

The correct answer is: UGA

Question **38**

Not answered

Marked out of 1.00

Flag question

You have just seen Mr Perker in your outpatient clinic. He's a 28-year-old man who uses cocaine in a dependent fashion, and whose life is severely impaired due to his drug use. The medical students in your clinic wish to better understand the regions of the brain associated with the reward system, implicated in the repetition of drug-taking behaviours. You explain that the region involved is:

Select one:

- Suprachiasmatic nucleus
- The ventral tegmental area
- The vermis cerebelli
- The posterior pituitary
- The paraventricular nucleus of the hypothalamus

Your answer is incorrect.

The mesolimbic dopamine pathway plays a central role in the reward system, connecting the ventral tegmental area with the nucleus accumbens (located in the ventral striatum).

The correct answer is: The ventral tegmental area

Question **39**

Not answered

Marked out of 1.00

Flag question

In which of the following types of aphasia is speech fluent, incomprehensible, with preserved repetition?

Select one:

- Transcortical motor aphasia
- Wernicke's aphasia
- Broca's aphasia
- Conduction aphasia
- Transcortical sensory aphasia

Your answer is incorrect.

Transcortical sensory aphasia involves damage to specific areas of the temporal lobe of the brain, resulting in symptoms such as poor auditory comprehension, relatively intact repetition, and fluent speech with semantic paraphasias present. It is a fluent aphasia similar to Wernicke's, with the exception of a strong ability to repeat words and phrases. In Broca's aphasia individuals have great difficulty forming complete sentences with generally only basic content words (leaving out words like "is" and "the"). In conduction aphasia individual can comprehend what is being said and is fluent in spontaneous speech, but they cannot repeat what is being said to them. Transcortical motor aphasia is characterized by non-fluent speech output, with good comprehension and repetition.

The correct answer is: Transcortical sensory aphasia

Question **40**

Not answered

Marked out of 1.00

Flag question

Miriam Lossef was involved in a sporting accident resulting in damage to her frontal lobe. It seems as though the other areas of the brain remained intact. You conduct a number of tests to determine the extent of the damage. Which of the following is a localising sign for frontal lobe damage?

Select one:

- Tinnitus
- Unilateral anosmia

- Impairment in propioception
- Dysgraphia
- Sensory aphasia

Your answer is incorrect.

A and D are parietal lobe signs, B and D are temporal lobe signs

The correct answer is: Unilateral anosmia

Question **41**

Not answered

Marked out of 1.00

Flag question

Mrs June Murphy is a 69-year-old woman who has presented to her GP with difficulty swallowing and a change in her voice. She is diagnosed with bulbar palsy following an MRI. Which of the following is she most likely to have?

Select one:

- Ophthalmoplegia
- Emotional Lability
- Upper motor neuron lesions
- Wasting of tongue muscles
- Preserved Jaw Jerk Reflex

Your answer is incorrect.

Bulbar palsy refers to a range of different signs and symptoms linked to impairment of function of the cranial nerves 9, 10, 11 and 12 which occurs due to a lower motor neuron lesion in the medulla oblongata or from lesions of the lower cranial nerves outside the brainstem. The eyes are not typically affected and the jaw jerk is absent or reduced. The tongue is commonly atrophic and shows fasciculations.

The correct answer is: Wasting of tongue muscles

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